

Course 6044B

Semester VI

Theory

4 Credits

Bio-medical Electronics

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes

Electronics Engineering

Contact hours

60

Teaching scheme

3:1:0:0

Credits

4

CIE / ESE

Theory CIE 40 / Theory ESE 60

Self-learning

5.5 h/week

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 6044B as Bio-medical Electronics in Semester VI for Electronics Engineering. The official SITTTTR detailed course page was checked at the indexed link and returned “Requested file not found.” With the user's approved public-source approach, this handbook is transparently based on NPTEL IIT Kharagpur’s Biomedical Signal Processing course and polytechnic medical electronics curricula. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Medical designs, bio-signal acquisitions, therapeutic protocols and imaging systems must be based on approved engineering plans, certified medical data, current safety standards and competent professional review.

Verified source note: Verified sources support bio-potentials, medical amplifiers, therapeutic devices, and patient safety. The four-module sequence is a learning path, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60

Contact hours

4

Credits

Theory CIE 40

CIE

Theory ESE 60

ESE

Course introduction

Bio-medical Electronics studies the principles, devices and systems used for acquiring, processing and monitoring biological signals from the human body. It develops the ability to analyze bio-potentials like ECG and EEG, evaluate medical amplifiers and imaging systems, understand therapeutic devices like pacemakers, and coordinate patient-safe electronic solutions while respecting the technical and ethical boundaries of electronic and biomedical engineering.

Malayalam support: □ handbook □□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□□; □□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□□□.

Prerequisite foundation

Basic electronics, operational amplifiers, digital electronics and human anatomy basics.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 • Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 • Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 • Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 • Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain the origin of bio-potentials and the characteristics of various bio-electrodes.
2. Analyze the acquisition and amplification of bio-signals including ECG, EEG and EMG.
3. Evaluate the operation and clinical applications of diagnostic and therapeutic medical equipment.
4. Explain the systematic safety standards, imaging systems and future trends in bio-medical electronics.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

Malayalam support: CO കണ്ടെത്തുക exam-ൽ ഉത്തരം നൽകുന്നതിനായി ഉത്തരം നൽകുക. Define, explain, apply എന്നിവ action words ഉപയോഗിക്കുക.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain the origin of bio-potentials and characteristics of bio-electrodes.	Understand
CO2	Analyze the acquisition and processing of bio-signals like ECG and EEG.	Apply
CO3	Evaluate the operation of therapeutic and diagnostic medical equipment.	Apply
CO4	Explain medical imaging systems and patient safety standards.	Understand

Official CO-PO articulation matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7-PO11
CO1	3	2	2	2	2	-	-
CO2	3	2	2	2	2	-	-
CO3	3	2	2	2	2	-	-
CO4	3	2	2	2	2	-	-

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters

Module	Title	Official major topics	Hours	CO	Handbook section
1	Bio-potentials and Electrodes	Origin of bio-potentials: Resting and Action potential; Sodium-Potassium pump; Electrode-electrolyte interface; Polarization; Types of electrodes: Surface, Needle, Microelectrodes.	Approx. 14 hours	CO1	Module 1
2	Bio-signal Acquisition and Amplifiers	ECG, EEG, EMG, ERG: Principles, Lead systems and clinical significance; Bio-amplifiers: Differential, Instrumentation, Isolation, and Chopper amplifiers; Signal conditioning basics.	Approx. 16 hours	CO2	Module 2
3	Diagnostic and Therapeutic Equipment	Patient monitoring systems: Heart rate, Respiration, Temperature; Therapeutic devices: Pacemakers (Internal/External), Defibrillators (AC/DC), Ventilators; Short-wave Diathermy.	Approx. 16 hours	CO3	Module 3
4	Medical Imaging and Patient Safety	X-ray machine: Block diagram and operation; Computed Tomography (CT) basics; MRI and Ultrasound imaging; Patient safety: Macroshock, Microshock, Leakage current; IEC 60601 standards.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Bio-potentials and Electrodes

Origin of bio-potentials: Resting and Action potential; Sodium-Potassium pump; Electrode-electrolyte interface; Polarization; Types of electrodes: Surface, Needle, Microelectrodes.

CO1 · Approx. 14 hours

Bio-signal Acquisition and Amplifiers

ECG, EEG, EMG, ERG: Principles, Lead systems and clinical significance; Bio-amplifiers: Differential, Instrumentation, Isolation, and Chopper amplifiers; Signal conditioning basics.

CO2 · Approx. 16 hours

Diagnostic and Therapeutic Equipment

Patient monitoring systems: Heart rate, Respiration, Temperature; Therapeutic devices: Pacemakers (Internal/External), Defibrillators (AC/DC), Ventilators; Short-wave Diathermy.

CO3 · Approx. 16 hours

Medical Imaging and Patient Safety

X-ray machine: Block diagram and operation; Computed Tomography (CT) basics; MRI and Ultrasound imaging; Patient safety: Macroshock, Microshock, Leakage current; IEC 60601 standards.

CO4 · Approx. 14 hours

MODULE 1

Bio-potentials and Electrodes

Origin of bio-potentials: Resting and Action potential; Sodium-Potassium pump; Electrode-electrolyte interface; Polarization; Types of electrodes: Surface, Needle, Microelectrodes.

Approx. 14 hours

CO1

Understand · Apply

Learning goals

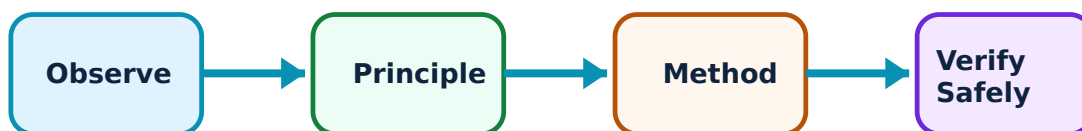
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Bio-potentials and Electrodes: identify the system, state its principle, follow the correct method and verify the result safely.

1. Bio-potential Fundamentals–

Bio-potentials are electrical signals generated by biological cells (nerve, muscle). The resting potential is maintained by the sodium-potassium pump. When stimulated, the cell depolarizes, creating an action potential. These signals are the basis for ECG, EEG, and EMG.

Study and application method

Describe the 'Sodium-Potassium Pump' conceptually; explain the 'Depolarization' and 'Repolarization' phases of an action potential.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the 'Sodium-Potassium Pump' conceptually; explain the 'Depolarization' and 'Repolarization' phases of an action potential.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ഏതു concept ഈശ്വരം ഈശ്വരം ഈശ്വരം. ഈശ്വരം diagram / circuit / formula / procedure ഈശ്വരം ഈശ്വരം. ഈശ്വരം result ഈശ്വരം application ഈശ്വരം ഈശ്വരം ഈശ്വരം ഈശ്വരം. Technical terms English- ഈശ്വരം ഈശ്വരം.

Common mistake / precaution: Bio-potentials are in the microvolt to millivolt range. Do not use standard electronic probes for bio-signal acquisition without authorised high-impedance, low-noise bio-electrodes.

2. Bio-electrodes–

Electrodes convert ionic current in the body to electronic current in the lead wires. Surface electrodes (metal plate, suction cup) are used for non-invasive recording. Needle and microelectrodes are used for invasive or cellular-level measurements.

Study and application method

Compare 'Surface' and 'Needle' electrodes in a conceptual table; explain the role of 'Electrode Jelly' in reducing skin-electrode impedance.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare 'Surface' and 'Needle' electrodes in a conceptual table; explain the role of 'Electrode Jelly' in reducing skin-electrode impedance.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: കോൺസെപ്റ്റ് താരതമ്യം ചെയ്തുകൊണ്ടും, ഡയഗ്രാം / സിരക്യൂട്ട് / ഫോർമുല / പ്രോസീഡർ ഉപയോഗിച്ചും ഉത്തരം നൽകണം. ഏതെങ്കിലും റിസൾട്ട് ഉപയോഗിച്ച് അപ്ലിക്കേഷൻ വിശദീകരിക്കണം. Technical terms English-ൽ ഉപയോഗിക്കണം.

Common mistake / precaution: Improper electrode contact can cause artifacts or skin irritation. All electrode applications must follow authorised hygiene and skin-preparation protocols.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Bio-signal Acquisition and Amplifiers

ECG, EEG, EMG, ERG: Principles, Lead systems and clinical significance; Bio-amplifiers: Differential, Instrumentation, Isolation, and Chopper amplifiers; Signal conditioning basics.

Approx. 16 hours

CO2

Understand · Apply

Learning goals

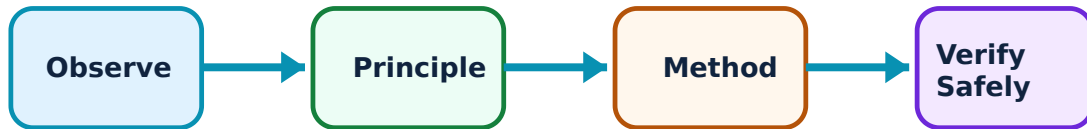
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Bio-signal Acquisition and Amplifiers: identify the system, state its principle, follow the correct method and verify the result safely.

1. Bio-signal Characteristics–

ECG (Electrocardiogram) records heart activity. EEG (Electroencephalogram) records brain waves. EMG (Electromyogram) records muscle activity. Each signal has specific frequency and amplitude ranges, requiring tailored acquisition systems.

Study and application method

Sketch a typical 'ECG Waveform' and label the P, QRS, and T components; list the frequency range of 'EEG' signals.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch a typical 'ECG Waveform' and label the P, QRS, and T components; list the frequency range of 'EEG' signals.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ഏതു concept ഈശ്വരം ഈശ്വരം ഈശ്വരം. ഈശ്വരം diagram / circuit / formula / procedure ഈശ്വരം ഈശ്വരം. ഈശ്വരം result ഈശ്വരം application ഈശ്വരം ഈശ്വരം ഈശ്വരം ഈശ്വരം. Technical terms English- ഈശ്വരം ഈശ്വരം.

Common mistake / precaution: Bio-signals are prone to 50/60Hz power line interference. All acquisition systems must include authorised notch filters and right-leg drive (RLD) circuits for common-mode rejection.

2. Medical Amplifiers–

Bio-amplifiers must have high input impedance, high CMRR, and low noise. Instrumentation amplifiers provide precise gain. Isolation amplifiers protect the patient from leakage current by providing electrical isolation between the body and the mains-powered equipment.

Study and application method

Explain the importance of 'High CMRR' in bio-amplifiers; describe the working of an 'Isolation Amplifier' conceptually.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain the importance of 'High CMRR' in bio-amplifiers; describe the working of an 'Isolation Amplifier' conceptually.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: കോൺസെപ്റ്റ് വ്യക്തമായി വിശദീകരിക്കുക. വോൾട്ട്, കറന്റ്, ഫ്രീക്വൻസി, ടൈം എന്നിവയുടെ യൂണിറ്റ് ഉൾപ്പെടെയുള്ളവയെക്കുറിച്ച്. വോൾട്ട്, കറന്റ്, ഫ്രീക്വൻസി, ടൈം എന്നിവയുടെ യൂണിറ്റ് ഉൾപ്പെടെയുള്ളവയെക്കുറിച്ച്. diagram / circuit / formula / procedure വ്യക്തമായി വിശദീകരിക്കുക. വോൾട്ട്, കറന്റ്, ഫ്രീക്വൻസി, ടൈം എന്നിവയുടെ യൂണിറ്റ് ഉൾപ്പെടെയുള്ളവയെക്കുറിച്ച്. result വ്യക്തമായി വിശദീകരിക്കുക. application വ്യക്തമായി വിശദീകരിക്കുക. Technical terms English-ൽ വ്യക്തമായി വിശദീകരിക്കുക.

Common mistake / precaution: Amplifier saturation can mask critical bio-signal features. Do not use high-gain amplifiers without authorised baseline-restoration and AC-coupling circuits.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Diagnostic and Therapeutic Equipment

Patient monitoring systems: Heart rate, Respiration, Temperature; Therapeutic devices: Pacemakers (Internal/External), Defibrillators (AC/DC), Ventilators; Short-wave Diathermy.

Approx. 16 hours

CO3

Understand · Apply

Learning goals

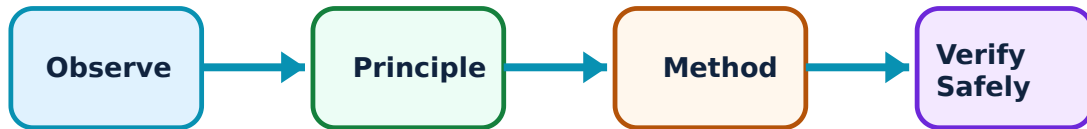
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Diagnostic and Therapeutic Equipment: identify the system, state its principle, follow the correct method and verify the result safely.

1. Patient Monitoring–

Bedside monitors continuously track vital signs. Heart rate is derived from ECG, respiration from thoracic impedance, and temperature from thermistors. Alarms are triggered if parameters exceed safe limits.

Study and application method

List the components of a 'Bedside Patient Monitor'; explain how 'Thoracic Impedance' is used to measure respiration rate.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: List the components of a 'Bedside Patient Monitor'; explain how 'Thoracic Impedance' is used to measure respiration rate.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ഏതു concept ഈ ചോദ്യം പരിഹരിക്കാൻ ഉപയോഗിക്കണം. ഈ diagram / circuit / formula / procedure എങ്ങനെ ഉപയോഗിക്കണം. ഈ result ഈ application ഈ ഏതു ഏതു ഉപയോഗിക്കണം. Technical terms English-ൽ എങ്ങനെ എഴുതണം.

Common mistake / precaution: False alarms in monitors can lead to 'alarm fatigue'. All monitoring systems must follow authorised alarm-threshold settings and sensor-integrity checks.

2. Therapeutic Devices–

Pacemakers provide electrical pulses to maintain heart rhythm. Defibrillators deliver a high-energy shock to restore normal heart rhythm during cardiac arrest. Ventilators assist or replace spontaneous breathing by delivering pressurized air/oxygen.

Study and application method

Differentiate between 'Internal' and 'External' pacemakers; describe the operation of a 'DC Defibrillator' conceptually.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Differentiate between 'Internal' and 'External' pacemakers; describe the operation of a 'DC Defibrillator' conceptually.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ഏതു concept ന്നുള്ള diagram / circuit / formula / procedure ന്നുള്ള result application ന്നുള്ള Technical terms English-ൽ എഴുതുക.

Common mistake / precaution: Defibrillators deliver lethal energy levels. Do not operate therapeutic devices without authorised medical supervision and adherence to equipment-specific safety protocols.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Medical Imaging and Patient Safety

X-ray machine: Block diagram and operation; Computed Tomography (CT) basics; MRI and Ultrasound imaging; Patient safety: Macroshock, Microshock, Leakage current; IEC 60601 standards.

Approx. 14 hours

CO4

Understand · Apply

Learning goals

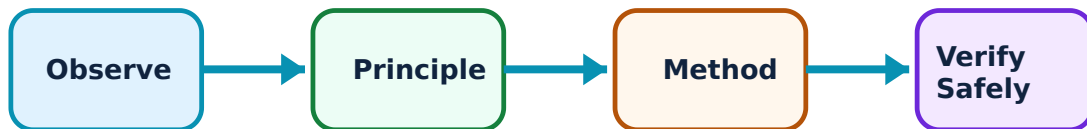
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Medical Imaging and Patient Safety: identify the system, state its principle, follow the correct method and verify the result safely.

1. Medical Imaging Systems–

Imaging systems allow non-invasive visualization of internal organs. X-rays use ionizing radiation. CT provides cross-sectional views. MRI uses magnetic fields and RF pulses. Ultrasound uses high-frequency sound waves, making it safe for fetal monitoring.

Study and application method

Sketch the block diagram of an 'X-ray Machine'; explain the principle of 'Ultrasound Imaging' conceptually.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch the block diagram of an 'X-ray Machine'; explain the principle of 'Ultrasound Imaging' conceptually.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ഏതു concept ഈശ്വരം ഈശ്വരം ഈശ്വരം. ഈശ്വരം diagram / circuit / formula / procedure ഈശ്വരം ഈശ്വരം. ഈശ്വരം result ഈശ്വരം application ഈശ്വരം ഈശ്വരം ഈശ്വരം ഈശ്വരം. Technical terms English- ഈശ്വരം ഈശ്വരം.

Common mistake / precaution: X-rays and CT scans involve ionizing radiation hazards. All imaging procedures must follow authorised radiation protection standards (ALARA principle).

2. Patient Safety and Standards–

Safety is paramount in medical electronics. Macroshock is caused by large currents through the body, while microshock occurs when current flows directly to the heart (e.g., via catheters). IEC 60601 is the international standard for medical electrical equipment safety.

Study and application method

Compare 'Macroshock' and 'Microshock' conceptually; list three ways to minimize 'Leakage Current' in medical devices.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare 'Macroshock' and 'Microshock' conceptually; list three ways to minimize 'Leakage Current' in medical devices.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ഏതു concept നോക്കുക. ഏതു diagram / circuit / formula / procedure ഉപയോഗിക്കുക. ഏതു result application ഉപയോഗിക്കുക. Technical terms English-ൽ എഴുതുക.

Common mistake / precaution: Medical equipment must be regularly tested for leakage current and grounding integrity. Do not use medical devices without authorised periodic safety audits as per IEC 60601 standards.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Bio-potentials and Electrodes

Use the concept flow to revise observation, principle, method and verification.

Module 2: Bio-signal Acquisition and Amplifiers

Use the concept flow to revise observation, principle, method and verification.

Module 3: Diagnostic and Therapeutic Equipment

Use the concept flow to revise observation, principle, method and verification.

Module 4: Medical Imaging and Patient Safety

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Compare ECG and EEG signals in a conceptual table showing amplitude and frequency differences.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Sketch the electrode placement for a 'Standard 12-lead ECG' conceptually.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draw the block diagram of a Patient Monitoring System showing the sensors and display.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Calculate the common-mode rejection ratio (CMRR) conceptually for a given bio-amplifier.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Identify a medical electronic device in a clinic (e.g., BP monitor, Pulse oximeter) and note its features.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Bio-potentials and Electrodes

Explain Bio-potential Fundamentals and state one correct method, application or precaution.—

Bio-potentials are electrical signals generated by biological cells (nerve, muscle). The resting potential is maintained by the sodium-potassium pump. When stimulated, the cell depolarizes, creating an action potential. These signals are the basis for ECG, EEG, and EMG.

Answer framework: Describe the 'Sodium-Potassium Pump' conceptually; explain the 'Depolarization' and 'Repolarization' phases of an action potential.

Important point: Bio-potentials are in the microvolt to millivolt range. Do not use standard electronic probes for bio-signal acquisition without authorised high-impedance, low-noise bio-electrodes.

Explain Bio-electrodes and state one correct method, application or precaution.—

Electrodes convert ionic current in the body to electronic current in the lead wires. Surface electrodes (metal plate, suction cup) are used for non-invasive recording. Needle and microelectrodes are used for invasive or cellular-level measurements.

Answer framework: Compare 'Surface' and 'Needle' electrodes in a conceptual table; explain the role of 'Electrode Jelly' in reducing skin-electrode impedance.

Important point: Improper electrode contact can cause artifacts or skin irritation. All electrode applications must follow authorised hygiene and skin-preparation protocols.

Module 2 — Bio-signal Acquisition and Amplifiers

Explain Bio-signal Characteristics and state one correct method, application or precaution.—

ECG (Electrocardiogram) records heart activity. EEG (Electroencephalogram) records brain waves. EMG (Electromyogram) records muscle activity. Each signal has specific frequency and amplitude ranges, requiring tailored acquisition systems.

Answer framework: Sketch a typical 'ECG Waveform' and label the P, QRS, and T components; list the frequency range of 'EEG' signals.

Important point: Bio-signals are prone to 50/60Hz power line interference. All acquisition systems must include authorised notch filters and right-leg drive (RLD) circuits for common-mode rejection.

Explain Medical Amplifiers and state one correct method, application or precaution.—

Bio-amplifiers must have high input impedance, high CMRR, and low noise. Instrumentation amplifiers provide precise gain. Isolation amplifiers protect the patient from leakage current by providing electrical isolation between the body and the mains-powered equipment.

Answer framework: Explain the importance of 'High CMRR' in bio-amplifiers; describe the working of an 'Isolation Amplifier' conceptually.

Important point: Amplifier saturation can mask critical bio-signal features. Do not use high-gain amplifiers without authorised baseline-restoration and AC-coupling circuits.

Module 3 — Diagnostic and Therapeutic Equipment

Explain Patient Monitoring and state one correct method, application or precaution.—

Bedside monitors continuously track vital signs. Heart rate is derived from ECG, respiration from thoracic impedance, and temperature from thermistors. Alarms are triggered if parameters exceed safe limits.

Answer framework: List the components of a 'Bedside Patient Monitor'; explain how 'Thoracic Impedance' is used to measure respiration rate.

Important point: False alarms in monitors can lead to 'alarm fatigue'. All monitoring systems must follow authorised alarm-threshold settings and sensor-integrity checks.

Explain Therapeutic Devices and state one correct method, application or precaution.—

Pacemakers provide electrical pulses to maintain heart rhythm. Defibrillators deliver a high-energy shock to restore normal heart rhythm during cardiac arrest. Ventilators assist or replace spontaneous breathing by delivering pressurized air/oxygen.

Answer framework: Differentiate between 'Internal' and 'External' pacemakers; describe the operation of a 'DC Defibrillator' conceptually.

Important point: Defibrillators deliver lethal energy levels. Do not operate therapeutic devices without authorised medical supervision and adherence to equipment-specific safety protocols.

Module 4 — Medical Imaging and Patient Safety

Explain Medical Imaging Systems and state one correct method, application or precaution.—

Imaging systems allow non-invasive visualization of internal organs. X-rays use ionizing radiation. CT provides cross-sectional views. MRI uses magnetic fields and RF pulses. Ultrasound uses high-frequency sound waves, making it safe for fetal monitoring.

Answer framework: Sketch the block diagram of an 'X-ray Machine'; explain the principle of 'Ultrasound Imaging' conceptually.

Important point: X-rays and CT scans involve ionizing radiation hazards. All imaging procedures must follow authorised radiation protection standards (ALARA principle).

Explain Patient Safety and Standards and state one correct method, application or precaution. –

Safety is paramount in medical electronics. Macroshock is caused by large currents through the body, while microshock occurs when current flows directly to the heart (e.g., via catheters). IEC 60601 is the international standard for medical electrical equipment safety.

Answer framework: Compare 'Macroshock' and 'Microshock' conceptually; list three ways to minimize 'Leakage Current' in medical devices.

Important point: Medical equipment must be regularly tested for leakage current and grounding integrity. Do not use medical devices without authorised periodic safety audits as per IEC 60601 standards.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Bio-potentials and Electrodes.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Bio-signal Acquisition and Amplifiers.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Diagnostic and Therapeutic Equipment.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Medical Imaging and Patient Safety.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Origin of bio-potentials: Resting and Action potential; Sodium-Potassium pump; Electrode-electrolyte interface; Polarization; Types of electrodes: Surface, Needle, Microelectrodes..
2. Develop a complete answer or practical plan for: ECG, EEG, EMG, ERG: Principles, Lead systems and clinical significance; Bio-amplifiers: Differential, Instrumentation, Isolation, and Chopper amplifiers; Signal conditioning basics..
3. Develop a complete answer or practical plan for: Patient monitoring systems: Heart rate, Respiration, Temperature; Therapeutic devices: Pacemakers (Internal/External), Defibrillators (AC/DC), Ventilators; Short-wave Diathermy..
4. Develop a complete answer or practical plan for: X-ray machine: Block diagram and operation; Computed Tomography (CT) basics; MRI and Ultrasound imaging; Patient safety: Macroshock, Microshock, Leakage current; IEC 60601 standards..

LAST-MILE REVISION

Quick Revision

Module 1: Bio-potentials and Electrodes

Origin of bio-potentials: Resting and Action potential; Sodium-Potassium pump; Electrode-electrolyte interface; Polarization; Types of electrodes: Surface, Needle, Microelectrodes.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Bio-signal Acquisition and Amplifiers

ECG, EEG, EMG, ERG: Principles, Lead systems and clinical significance; Bio-amplifiers: Differential, Instrumentation, Isolation, and Chopper amplifiers; Signal conditioning basics.

- Match each topic to CO2.
- Redraw or rehearse one key method.

- Check one common mistake/precaution.

Module 3: Diagnostic and Therapeutic Equipment

Patient monitoring systems: Heart rate, Respiration, Temperature; Therapeutic devices: Pacemakers (Internal/External), Defibrillators (AC/DC), Ventilators; Short-wave Diathermy.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Medical Imaging and Patient Safety

X-ray machine: Block diagram and operation; Computed Tomography (CT) basics; MRI and Ultrasound imaging; Patient safety: Macroshock, Microshock, Leakage current; IEC 60601 standards.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus

Term / topic	One-line meaning
Bio-potential Fundamentals	Bio-potentials are electrical signals generated by biological cells (nerve, muscle).
Bio-electrodes	Electrodes convert ionic current in the body to electronic current in the lead wires.
Bio-signal Characteristics	ECG (Electrocardiogram) records heart activity.
Medical Amplifiers	Bio-amplifiers must have high input impedance, high CMRR, and low noise.
Patient Monitoring	Bedside monitors continuously track vital signs.
Therapeutic Devices	Pacemakers provide electrical pulses to maintain heart rhythm.
Medical Imaging Systems	Imaging systems allow non-invasive visualization of internal organs.
Patient Safety and Standards	Safety is paramount in medical electronics.

LEARNING RESOURCES

References

Recommended bio-medical electronics references

1. L. Cromwell, F. J. Weibell and E. A. Pfeiffer, Biomedical Instrumentation and Measurements.
2. R. S. Khandpur, Handbook of Biomedical Instrumentation.
3. John G. Webster, Medical Instrumentation: Application and Design.
4. Joseph J. Carr and John M. Brown, Introduction to Biomedical Equipment Technology.

Approved public bio-medical electronics sources

- <https://nptel.ac.in/courses/108105101>
- https://www.youtube.com/playlist?list=PLyqSpQzTE6M_rQyhzfm2XMkkFmwzDKKEo

These sources provide the verified study basis while official Course 6044B detail is unavailable; they do not replace official clinical designs, authorised medical device maintenance, certified equipment specifications or authorised hospital safety protocols.